

Aishwarya Ramanath Shanbhag

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EDUCATION

San Jose State University

Master of Science in Computer Science (In Progress – Semester 1)

Relevant Coursework: Topics in Machine Learning, Database Systems Principles

San Jose, CA

Aug 2026 – May 2028

PES University

Bachelor of Technology in Computer Science & Engineering; GPA: 3.6/4.0

Relevant Coursework: Machine Intelligence, Natural Language Processing, Big Data, Data Analytics

Bengaluru, India

Aug 2018 – May 2022

TECHNICAL SKILLS

Programming Languages: Python (Pandas, NumPy, Scikit-learn), SQL, R

ML & Statistical Modeling: Supervised & Unsupervised Learning, Predictive Modeling, Feature Engineering, Ensemble Techniques (XGBoost), Anomaly Detection, NLP, A/B Testing, Model Validation

Risk & Compliance Analytics: Fraud Detection, Financial Crime Risk Rating, AML/KYC, Regulatory Compliance, MRM

Data Engineering, Cloud & BI: ETL Pipelines, Spark, Snowflake, AWS, SageMaker, QuickSight, Tableau, Git, GitHub

WORK EXPERIENCE

NatWest Group

Data Scientist – Machine Learning, Fraud Detection & Prevention Squad

Bengaluru, India

Jun 2025 – Jul 2026

- Optimized and modified the **TMX ThreatMetrix** onboarding classification model, deploying algorithmic updates that prevented **\$94,000 in monthly fraud losses**, reduced **false positives by 58%**, and increased **fraud capture rates by 143%**.
- Designed **model monitoring dashboards** in **Amazon QuickSight**, embedding financial KPIs and **feature distribution shifts** for continuous algorithmic stability.
- Executed **model evaluations**, **silent period testing**, and **bias analyses** pre-deployment to assess **model drift** and fairness, partnering with cross-functional stakeholders on fraud strategy.

Societe Generale

Associate Data Scientist – Compliance Data Science & Financial Crime Client Risk Rating

Bengaluru, India

Aug 2022 – May 2025

- Led validation of **FCCR V3 Ongoing Monitoring models** as part of **Model Risk Management (MRM)** review, ensuring regulatory compliance and algorithmic integrity for a portfolio of **15 million+ retail banking clients**.
- Engineered **Nested Dichotomized Logit** and **Boosted C5.0** models, executing **3 monitoring cycles** across **8+ KPIs** including **PSI** to catch **data drift**, contributing to a **EUR 2 Million reduction** in regulatory penalties.
- Built a **BERTopic NLP pipeline** to automate text mining for **KYC** client and business comment analysis, eliminating **2 FTEs** of manual review.

Hewlett Packard Enterprise (HPE)

OCA Tool Developer Intern

Chennai, India

Feb 2022 – Aug 2022

- Shipped new platform features for One Config Advanced (OCA), including **search, cloning, and reusable configuration templates**, cutting **configuration time by 25%** for end users.
- Built **breakout, filtering, and attribute-selection modules** from the ground up, designing and testing core logic to increase platform productivity.
- Integrated and validated new product configurations, including **Aruba-specific settings**, expanding platform functionality through rigorous cross-team testing.

PROJECTS

Intrusion Detection Using Ensemble Techniques (Python, XGBoost, Ensemble Learning) Aug 2021 – Jan 2022

- Served as **Team Lead** for an **ensemble-based security analytics platform**, delivering a **99.93% risk detection accuracy rate** and an ultra-low **0.07% false positive rate**.
- Combined a **Maximum Voting Ensemble** with **Gaussian Naive Bayes**, **Decision Trees**, and **XGBoost**, directing **feature selection** and **hyperparameter tuning** for enterprise-level scalability.
- Benchmarked ensemble performance against individual base classifiers, demonstrating consistent accuracy gains from the ensemble approach.

Social Distance Detection using YOLOv3 (Python, OpenCV, Deep Learning)

Jan 2021 – May 2021

- Built a real-time social distancing system with **YOLOv3** and **SSD MobileNet** detection models via a **Streamlit** interface.
- Applied **perspective transformation** for bird's-eye view mapping and calibrated **Euclidean distance** to classify individuals into high, low, and no-risk zones.
- Achieved **92% detection accuracy** (95% with transfer learning) and **95% tracking accuracy** for risk classification.